

Assignment 2 SECPH 2523

QUESTION 1

Table 2.1 STUDENTFINE records data on traffic offences by students in a college.

TABLE 2.1 STUDENTFINE							
StudentID	Name	LicenseNo	CarReg	ReceiptNo	Date	OffenceCode	Fine(RM)
F1011	CK Fong	P100	JGK9201	00-345	10/03/06	2	30.00
				00-690	15/03/06	1	20.00
				00-810	20/03/06	2	30.00
L782	AdamJames	P290	JJL99	00-459	13/03/06	1	20.00
H762	MalaDewi	P764	JJK7654	00-564	12/03/06	3	40.00
				00-770	19/06/06	2	30.00

a) Table 2.1 is said to be unnormalized table. Why?

Some cells have multiple value.

b) Transform the unnormalized Table 2.1 into a First Normal Form (1NF) table. Show the transformed 1 NF table. What is the primary key for this new 1 NF table.

Primary key is StudentID.

StudentID	Name	LicenseNo	CarReg	ReceiptNo	Date	OffenceCode	Fine(RM)
F1011	CK Fong	P100	JGK9201	00-345	10/03/06	2	30.00
F1011	CK Fong	P100	JGK9201	00-690	15/03/06	1	20.00
F1011	CK Fong	P100	JGK9201	00-810	20/03/06	2	30.00
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QUESTION 2

The following table is un-normalized table that is created to manage a consulting company's operations. Create its normalized forms (1NF, 2NF, and 3NF).

Note: Define the PK and FK for each table:

Project number	Project name	Employee number	Employee name	Rate category	Hourly rate
1023	Madagascar travel site	11	Vincent Radebe	A	\$60
		12	Pauline James	B	\$50
		16	Charles Ramoraz	C	\$40
1056	Online estate agency	11	Vincent Radebe	A	\$60
		17	Monique Williams	B	\$50

1 NF

Project number	Project name	Employee number	Employee name	Rate category	Hourly rate
1023	Madagascar travel site	11	Vincent Radebe	A	\$60
1023	Madagascar travel site	12	Pauline James	B	\$50
1023	Madagascar travel site	16	Charles Romaraz	C	\$40
1056	Online estate agency	11	Vincent Radebe	A	\$60
1056	Online estate agency	17	Monique Williams	B	\$50

Primary key is Project number.

2 NF

Project table

Project number	Project name
1023	Madagascar travel site
1056	Online estate agency

Primary key is Project number.

Employee table

Employee number	Employee name	Rate category	Hourly rate
11	Vincent Radebe	A	\$60
12	Pauline James	B	\$50
16	Charles Romaraz	C	\$40
17	Monique Williams	B	\$50

Primary key is Employee number.

3 NF

Project table

Project number	Project name
1023	Madagascar travel site
1056	Online estate agency

Primary key is Project number.

Employee table

Employee number	Employee name
11	Vincent Radebe
12	Pauline James
16	Charles Romaraz
17	Monique Williams

Primary key is Employee number.

Rate table

Rate category	Hourly rate
A	\$60
B	\$50
C	\$40

Primary key is Rate category.

Assignment table

Project number	Employee number	Rate category
1023	11	A
1023	12	B
1023	16	C
1056	11	A
1056	17	B

Foreign key is Project number references Project table, Employee number reference Employee table, Rate category reference Rate table.